

Relate Fractions, Decimals, and Percents

Lesson 4-2

Name: _____

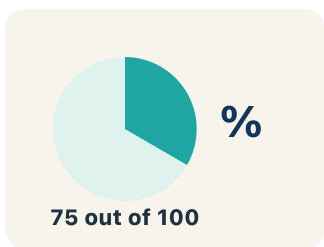
Date: _____

Class: _____

Key Vocabulary

Level 1 support

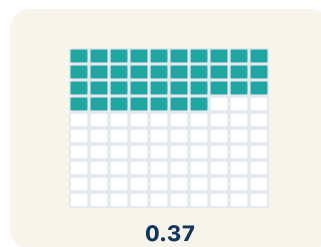
Picture first, then the word, then a plain-language meaning. Say each word out loud.



25% means 25 out of 100

Percent

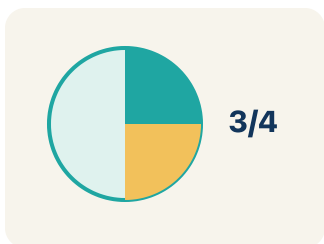
A way to compare a number to 100, shown with the % sign.



0.75 means seventy-five hundredths

Decimal

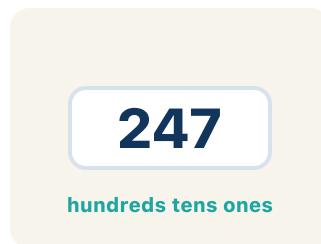
A number with a dot, like 0.5, that shows a part.



$\frac{1}{2} = 0.5 = 50\%$

Equivalent

Having the same value, just written a different way.

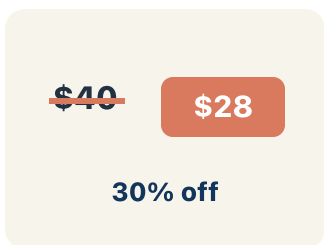


hundreds tens ones

Common benchmarks: 25%, 50%, 75%

Benchmark

A familiar number you use to guess, like 50% or $\frac{1}{2}$.



\$40 shirt at 25% off: discount = \$10

Discount

Money taken off the first price to make it cheaper.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can write equivalent fractions, decimals, and percents for the same value.

1 Notice

You're designing the scoring system for a new arcade game.

2 Key idea

I can write equivalent fractions, decimals, and percents for the same value.

3 Words I'll use

Percent

Decimal

Equivalent

Benchmark

Discount



Think About It

- What are three different ways to write the same number?
- If a player scored $\frac{3}{4}$ of the points, what percent is that?
- Are some fractions easier to convert than others?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: Convert $\frac{3}{5}$ to a percent.

A. 60%

B. 35%

C. 53%

D. 65%

Step 1

$$3 \div 5 = 0.6 = 60\%.$$



Answer: A. 60%

 We do — together

Solve this one with your class using the same steps.

Problem: Convert $\frac{7}{20}$ to a percent.

- A. 35%
- B. 70%
- C. 20%
- D. 14%

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Which decimal is equivalent to 45%?

- A. 0.45
- B. 4.5
- C. 0.045
- D. 45.0

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

A player scored $\frac{3}{4}$ of the points. The leaderboard shows percents and the game engine uses decimals. What are three different ways to write the same score?

Level 1 support

Start here: $\frac{3}{4}$, 0.75, and 75% all name the same score.

 **Need a hint?**

Sentence starters (optional)

The fraction ____ is the same as the decimal ____ and the percent ____.

La fracción ____ es igual al decimal ____ y al porcentaje ____.

I can write the same value as a fraction, a ____, or a ____.

Puedo escribir el mismo valor como fracción, ____ o ____.

Word bank: percent decimal equivalent benchmark

Listen for: Students give $\frac{3}{4} = 0.75 = 75\%$ and recognize these are three forms of one value, not three different scores.

Level 2

Which form would be easiest to read on a leaderboard, and which is easiest for the game engine to calculate with? Justify your choices.

- The ____ form is easiest to read because ____.
- The ____ form is easiest to calculate with because ____.

EXPLORE

How did you convert a fraction like $\frac{3}{5}$ into a decimal and a percent in the table?

Level 1 support

Start here: I divided the numerator by the denominator, then multiplied by 100 to get the percent.



Need a hint?

Sentence starters (optional)

To change ____ to a decimal I divided ____ by ____.

Para cambiar ____ a decimal dividí ____ entre ____.

To get the percent I multiplied the decimal by ____.

Para obtener el porcentaje multipliqué el decimal por ____.

Word bank: percent decimal equivalent benchmark

Listen for: Students describe dividing numerator by denominator ($3 \div 5 = 0.6$) and multiplying by 100 to get 60%, and call the three forms equivalent.

Level 2

Some fractions like $\frac{1}{4}$ give a clean decimal, but others do not. How can a benchmark like $\frac{1}{2}$ or $\frac{1}{4}$ help you estimate before you divide?

- A benchmark helps because ____.
- I know ____ is close to ____ because ____.

CONNECT

A sale says 'Everything $\frac{3}{5}$ off!' One friend says that is 35% off and another says 60% off. Who is right, and how do you know?

Level 1 support

Start here: $\frac{3}{5}$ equals 60%, so the friend who said 60% off is correct.

 Need a hint?

Sentence starters (optional)

The friend who said ____% is correct because $\frac{3}{5} = \underline{\hspace{1cm}} = \underline{\hspace{1cm}}\%$.

El amigo que dijo ____% tiene razón porque $\frac{3}{5} = \underline{\hspace{1cm}} = \underline{\hspace{1cm}}\%$.

The other friend is wrong because ____.

El otro amigo está equivocado porque ____.

Word bank: percent decimal equivalent discount

Listen for: Students convert $\frac{3}{5}$ to 0.6 and 60%, identify 60% off as correct, and can find the \$40 game becomes \$16.

Level 2

The other friend got 35% by reading the digits 3 and 5. Explain why that shortcut fails, and describe a discount where reading the digits would accidentally give the right percent.

- Reading the digits fails because ____.
- It would only work by accident when ____.

Write About the Math

The Writing Revolution

I can explain my conversions using the words percent, decimal, equivalent, and benchmark.

1. Kernel Sentence subject + verb

Model: Percent is a way to compare a number to 100, shown with the % sign.

Porcentaje es una manera de comparar un número con 100, con el signo %.

Write a kernel sentence about percent. Use a subject and a verb.

Escribe una oración base sobre porcentaje. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Percent matters in math

Porcentaje importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Percent matters in math because ____.

Porcentaje importa en matemáticas porque ____.

but
pero

Percent matters in math, but ____.

Porcentaje importa en matemáticas, pero ____.

so
entonces

Percent matters in math, so ____.

Porcentaje importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about percent.
Di un hecho verdadero sobre percent.

Percent ____.

Question *Pregunta*

Ask a question about percent.
Haz una pregunta sobre percent.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about percent.
Muestra entusiasmo sobre percent.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with percent.
Dile a un compañero qué hacer con percent.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The fraction ____ is the same as the decimal ____ and the percent ____.

La fracción ____ es igual al decimal ____ y al porcentaje ____.

I can write the same value as a fraction, a ____, or a ____.

Puedo escribir el mismo valor como fracción, ____ o ____.

To change ____ to a decimal I divided ____ by ____.

Para cambiar ____ a decimal dividí ____ entre ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which is the largest value?

- A. 0.7
- B. 65%
- C. $\frac{3}{5}$
- D. 0.59

Show your work:

2. What is 25% as a fraction in simplest form?

- A. $\frac{1}{4}$
- B. $\frac{25}{10}$
- C. $\frac{1}{5}$
- D. $\frac{4}{1}$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student says $\frac{3}{8}$ is greater than 40% because 3 and 8 are bigger numbers than 4 and 0. Is the student correct? Convert both values to the same form and explain.

Sentence starter: The student is ____ because $\frac{3}{8} = \frac{\quad}{\quad} = \quad\%$, which is ____ than 40%.

Show your work:

Reflect — Exit Ticket

Which shows 0.45 as a fraction and a percent?

- A. $\frac{9}{20}$ and 45%
- B. $\frac{45}{10}$ and 4.5%
- C. $\frac{4}{5}$ and 45%
- D. $\frac{9}{20}$ and 4.5%

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $35\% - 7 \div 20 = 0.35 = 35\%$.
2. **You Try (notes):** A. 0.45 — *To convert a percent to a decimal, divide by 100: $45 \div 100 = 0.45$.*
3. **Try It 1:** A. $0.7 - 0.7 = 70\%$, larger than 65% , 60% ($3/5$), and 59% .
4. **Try It 2:** A. $1/4 - 25\% = 25/100 = 1/4$.
5. **Exit Ticket:** A. $9/20$ and $45\% - 0.45 = 45/100 = 9/20$ (simplified). $0.45 \times 100 = 45\%$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Percent is a way to compare a number to 100, shown with the % sign.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).